Discrete Vertical Hurwitz Calculus

Stieltjes jets, polylogarithmic sampling,
and resonant contact terms

Research continuation prepared for
Vladimir Reshetnikov's ProveIt project

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Abstract

We develop an exact calculus for sums over a vertical lattice in the Hurwitz argument. Euler–Maclaurin compensation and an elementary periodization argument give a normally convergent spectral identity whose right-hand side is a Lambert series of polylogarithms. Its complete Taylor expansion at order one evaluates every compensated generalized Stieltjes lattice sum, with finite Bell-polynomial coefficients. Horizontal unit differences eliminate the Lambert series and leave finitely many polylogarithms and their order derivatives. Anchored antiderivatives are obtained by raising the polylogarithm order.

At nonpositive integer orders we prove a parity-sensitive contact law. First spectral derivatives have absolutely convergent lattice sums at negative odd orders; at negative even orders the symmetric sum differs from the spectral continuation by an explicit Bernoulli constant. The Gamma case is an already published specialization of Arabi Ardehali and Rosengren, not a novelty claim. All higher, sufficiently compensated spectral jets are expressed by a finite compiler involving generalized harmonic numbers and zeta values. Mean-zero periodic lattice weights annihilate every first-jet contact.

A second theorem evaluates lattice products of arbitrary positive-order polygammas as a single Hurwitz term plus exponentially convergent image sums, each image being a finite Hurwitz–Lerch combination. We also give an all-index Stieltjes-difference pairing and an explicit four-shift digamma formula. Proofs distinguish ordinary absolute sums, symmetric limits, and analytic continuation. The package includes 268 exact algebraic assertions, 38 independent or cross-route numerical comparisons, two corruption controls, and a scoped source audit. The numerical tests are not interval certificates. No claim is made to settle the manuscript's S_6, revised S_8, or general unequal-tail prefix conjectures.

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1 Scope, source frontier, and contribution

The ProveIt manuscript has developed several complementary calculi: periodic Hurwitz correlations, spectral derivatives, finite parts, Gamma products, and harmonic-tail reconstruction. In particular, the inspected vertical-pairing report concerns ordinary integrals in the imaginary part of the Hurwitz argument, rather than discrete sampling [9]. The present article replaces that continuous vertical geometry by the lattice

$$a + \mathrm{i}h\mathbb{Z}, \qquad h > 0, \qquad \operatorname{Re} a > 0.$$

This change produces nonzero sampling images as well as an explicit boundary contact. It therefore requires its own convergence statements and cannot be obtained by simply replacing an integral sign by a sum.

The principal exact results are the compensated sampling theorem (Theorem 3.2), the complete Stieltjes coefficient identity (Theorem 4.1), the negative-integer contact law (Theorem 7.3), the harmonic-number jet compiler (Theorem 8.1), the finite-image polygamma pairing (Theorem 9.1), and the four-shift formula (Theorem 10.2). The proofs are independent of the pending harmonic-rank conjectures.

What is inherited. The Hurwitz continuation, Euler–Maclaurin expansion, Gamma special values, and polylogarithm/Lerch definitions are classical [1, 2, 3, 4]. Periodization is a standard form of Poisson summation [5]. Stieltjes summation and parameter differentiation also have substantial antecedents; Coffey's summation relations are one relevant example [6]. Most importantly, the Gamma product obtained below is equivalent to the published product formula of Arabi Ardehali and Rosengren [7, Eq. (13)]. That specialization, including its constant, is explicitly credited.

What is proposed for integration. The contribution is the unified discrete spectral calculus, its all-index convergent Stieltjes and higher-resonance consequences, and the explicit image formulas for products. These are proved here as exact identities. They are proposed continuations relative to the inspected repository sources, not assertions of exhaustive historical priority. The distinction matters: a preliminary reconstruction calculation in this investigation reproduced the already proved smaller-tail-order-two family in a later repository report [10]. It is not included as a new result.

Audit limits. The main branch changed during inspection. Accordingly, the audit records individual source paths and blob identifiers rather than claiming a single immutable snapshot for all reads. The incoming directory was listed and its intake instructions were read, but the interiors of its five then-listed ZIP archives were not available for inspection in this run. No complete audit of those archives, of all manuscript volumes, or of the literature is claimed. No repository file was modified. Section 11 records the concrete correction and the distinction between that correction and a mathematical warning.

2 Conventions and the compensated Mellin family

Fix $h > 0$ and write

$$\tau = \frac{2\pi}{h}, \qquad q = e^{-\tau}, \qquad z = e^{-\tau a}, \qquad w_n = a + \mathrm{i}hn \quad (n \in \mathbb{Z}). \tag{2.1}$$

Initially $\operatorname{Re} a > 0$; hence $|z| < 1$. Every power of w uses the analytic logarithm $\operatorname{Log} w$ in the right half-plane. The analytic branch of $\log \Gamma(w)$ is real on the positive real axis. It is not the principal logarithm of a numerically computed complex Gamma value.

We use the Laurent convention

$$\zeta(1+\varepsilon,w)=\frac{1}{\varepsilon}+\sum_{m=0}^{\infty}\frac{(-1)^m}{m!}\gamma_m(w)\varepsilon^m,\qquad \gamma_0(w)=-\psi(w). \tag{2.2}$$

Here $\psi=\Gamma'/\Gamma$, $(s)_k=s(s+1)\cdots(s+k-1)$, B_{2j} are Bernoulli numbers, and

$$H_N^{(r)}=\sum_{k=1}^{N}k^{-r},\qquad H_N=H_N^{(1)},\qquad H_0^{(r)}=0.$$

The Bernoulli convention is $B_1=-1/2$; no formula below depends on a software package's alternative value for B_1.

Definition 2.1 (Compensation)**.** For an integer $M\ge 0$, put

$$C_M(t)=\frac{1}{1-e^{-t}}-\frac{1}{t}-\frac{1}{2}-\sum_{j=1}^{M}\frac{B_{2j}}{(2j)!}t^{2j-1}, \tag{2.3}$$

$$\mathcal{R}_M(s,w)=\zeta(s,w)-\frac{w^{1-s}}{s-1}-\frac{1}{2}w^{-s}-\sum_{j=1}^{M}\frac{B_{2j}}{(2j)!}(s)_{2j-1}w^{1-s-2j}. \tag{2.4}$$

The singularity at $s=1$ in the combined expression is filled by its removable limit. Thus $\mathcal{R}_M$ is entire in s for each $\operatorname{Re} w>0$.

The subtractions are fixed, explicit functions. No unspecified regularization or finite-part convention is hidden in (2.4). Near zero,

$$C_M(t)=\frac{B_{2M+2}}{(2M+2)!}t^{2M+1}+O(t^{2M+3}). \tag{2.5}$$

For $\operatorname{Re} s>-2M-1$, the Mellin identity is

$$\mathcal{R}_M(s,w)=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}e^{-wt}C_M(t)\,\mathrm{d}t. \tag{2.6}$$

It follows initially from the ordinary Hurwitz integral in a common convergence domain, by integrating the finitely many subtractions. Both sides are holomorphic on the indicated half-plane after removable values are filled, so the identity continues there. The factor $1/\Gamma(s)$ is entire; at its zeros the convergent integral is multiplied by zero.

Lemma 2.2 (Uniform vertical remainder)**.** *For s in a compact set and w in a fixed sector $|\arg w|\le\pi-\delta$, the Euler–Maclaurin expansion gives, for sufficiently large J,*

$$\mathcal{R}_M(s,w)=\sum_{j=M+1}^{J}\frac{B_{2j}}{(2j)!}(s)_{2j-1}w^{1-s-2j}+O(|w|^{-\operatorname{Re}s-2J-1}). \tag{2.7}$$

In particular, on compact subsets of $\{\operatorname{Re}s>-2M,\ \operatorname{Re}a>0\}$ the series

$$\sum_{n\in\mathbb{Z}}\partial_s^d\mathcal{R}_M(s,a+\mathrm{i}hn)$$

converges normally for each fixed $d\ge 0$.

Proof. Apply Euler–Maclaurin with additional terms to the defining Hurwitz sum; its remainder is an integral of a bounded periodic Bernoulli function times a power of $w+x$. In a closed sector avoiding the negative axis, this gives the stated uniform expansion. Equivalently, this is the standard sectorial Hurwitz expansion [1, §25.11(xii)], with the first M terms removed. On a vertical line the first remaining term has size $O(|n|^{-\operatorname{Re}s-2M-1})$. Spectral derivatives add only fixed powers of $\log(2+|n|)$, by Cauchy estimates on a slightly larger compact set. The exponent exceeds one uniformly on the indicated compacts. The finitely many small n terms are holomorphic and bounded there. □

Define the entire-in-order Lambert–polylogarithm family

$$\mathcal{L}_\nu(q,z)=\sum_{r=0}^{\infty}\operatorname{Li}_\nu(zq^r)=\sum_{k=1}^{\infty}\frac{z^k}{k^\nu(1-q^k)}. \tag{2.8}$$

Both expressions converge normally for $|z|<1$ and ν in compact subsets of $\mathbb{C}$. Every order derivative is obtained by inserting $(-\log k)^d$ in the second series. We will use

$$\begin{aligned}\mathcal{B}_M(s;a,\tau)=\mathcal{L}_{1-s}(q,z)-\frac{1}{2}\operatorname{Li}_{1-s}(z)-\frac{1}{\tau}\operatorname{Li}_{2-s}(z)\\ -\sum_{j=1}^{M}\frac{B_{2j}\tau^{2j-1}}{(2j)!}\operatorname{Li}_{2-s-2j}(z).\end{aligned} \tag{2.9}$$

It is entire in s. All τ-dependence, including q and z, is understood as in (2.1).

3 An exact vertical sampling theorem

Lemma 3.1 (Periodization with absolutely summable coefficients)**.** *Let $f\in L^1(\mathbb{R})$ be continuous and suppose $P(x)=\sum_{k\in\mathbb{Z}}f(x+\tau k)$ converges locally uniformly to a continuous τ-periodic function. If $\sum_n|\widehat{f}(hn)|<\infty$, with*

$$\widehat{f}(y)=\int_{\mathbb{R}}f(t)e^{-\mathrm{i}yt}\,\mathrm{d}t,\qquad h\tau=2\pi,$$

then

$$\sum_{n\in\mathbb{Z}}\widehat{f}(hn)=\tau\sum_{k\in\mathbb{Z}}f(\tau k). \tag{3.1}$$

Proof. The nth Fourier coefficient of P is $\widehat{f}(hn)/\tau$, by integration over one period and absolute integrability. Their absolute summability gives a uniformly convergent Fourier series. It represents P: the difference is a continuous periodic function with every Fourier coefficient zero, hence is zero, for example by Fejér approximation. Evaluate that Fourier series at $x=0$. □

Theorem 3.2 (Compensated vertical-lattice sampling)**.** *Let $M\ge 0$, $h>0$, $\operatorname{Re}a>0$, and $\operatorname{Re}s>-2M$. Then*

$$\boxed{\mathcal{S}_M(s;a,h):=\sum_{n\in\mathbb{Z}}\mathcal{R}_M(s,a+\mathrm{i}hn)=\frac{\tau^s}{\Gamma(s)}\mathcal{B}_M(s;a,\tau).} \tag{3.2}$$

The lattice sum is absolutely and normally convergent on this domain, including every fixed spectral derivative. The right side gives an entire continuation in s of this summed function.

Proof. Extend

$$f(t)=\frac{t^{s-1}e^{-at}C_M(t)}{\Gamma(s)}\quad(t>0)$$

by zero for $t\le 0$. At zero it is continuous and vanishes, because its leading power is t^{s+2M} and $\operatorname{Re}(s+2M)>0$. At positive infinity it decays exponentially times a fixed power, so its periodization is continuous and locally uniformly convergent. Its Fourier transform is $\mathcal{R}_M(s,a+\mathrm{i}y)$ by (2.6), and the Fourier coefficients are absolutely summable by Lemma 2.2. Thus

$$\mathcal{S}_M(s;a,h)=\frac{\tau^s}{\Gamma(s)}\sum_{k=1}^{\infty}k^{s-1}z^kC_M(\tau k).$$

Insert (2.3) and identify each of the absolutely convergent series. This gives (2.9) and (3.2). The normal convergence statement is Lemma 2.2; the continuation follows from the entire order dependence on the right. □

Remark 3.3 (Why the strict boundary matters). At $\operatorname{Re} s = -2M$, the density in the proof need not vanish at zero. The original lattice sum need not be absolutely convergent in a neighborhood of that order. Consequently, differentiating the entire right side there does not automatically differentiate an ordinary lattice sum. Section 7 computes the resulting contact exactly. This is a failure of interchange, not a failure of analytic continuation.

The theorem also covers an offset lattice $a + \mathrm{i}h(n+\theta)$: replace a by $a + \mathrm{i}h\theta$, so every polylogarithm argument is multiplied by $e^{-2\pi\mathrm{i}\theta}$. No new convergence issue arises.

4 Every generalized Stieltjes lattice sum

Set

$$K_m(w) = \gamma_m(w) + \frac{\operatorname{Log}^{m+1} w}{m+1} - \frac{\operatorname{Log}^m w}{2w}. \tag{4.1}$$

The exact Laurent cancellation gives

$$\mathcal{R}_0(1+\varepsilon, w) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} K_m(w)\varepsilon^m. \tag{4.2}$$

In particular, each $K_m(a + \mathrm{i}hn)$ is $O(\log^m(2+|n|)/(1+|n|)^2)$.

For a variable Y, define finite Bell polynomials $\mathcal{E}_j(Y)$ by

$$\exp\left(Yu + \sum_{r=2}^{\infty} \frac{(-1)^{r+1}\zeta(r)}{r} u^r\right) = \sum_{j=0}^{\infty} \mathcal{E}_j(Y)\frac{u^j}{j!}. \tag{4.3}$$

Only $r \le j$ contributes to $\mathcal{E}_j$. Their first values are

$$\begin{aligned} &\mathcal{E}_0(Y) = 1, && \mathcal{E}_1(Y) = Y, \\ &\mathcal{E}_2(Y) = Y^2 - \zeta(2), && \mathcal{E}_3(Y) = Y^3 - 3Y\zeta(2) + 2\zeta(3), \\ &\mathcal{E}_4(Y) = Y^4 - 6Y^2\zeta(2) + 8Y\zeta(3) + 3\zeta(2)^2 - 6\zeta(4). \end{aligned}$$

Write

$$\mathcal{Q}_\nu(a) = \mathcal{L}_\nu(q, z) - \frac{1}{2}\operatorname{Li}_\nu(z) - \frac{1}{\tau}\operatorname{Li}_{\nu+1}(z), \qquad L = \log\tau + \gamma. \tag{4.4}$$

Here γ is Euler's constant, and τ is held fixed in every a- or order derivative.

Theorem 4.1 (All-index Stieltjes identity)**.** *For $m \ge 0$ and $\operatorname{Re} a > 0$,*

$$\boxed{\sum_{n\in\mathbb{Z}} K_m(a + \mathrm{i}hn) = \tau \sum_{j=0}^{m} \binom{m}{j} (-1)^j \mathcal{E}_j(L)\, \partial_\nu^{m-j} \mathcal{Q}_\nu(a)\Big|_{\nu=0}.} \tag{4.5}$$

The sum is ordinary and absolutely convergent. There is no principal value and no finite part.

Proof. In (3.2), put $M = 0$ and $s = 1 + \varepsilon$. The reciprocal-Gamma expansion gives

$$\frac{\tau^{1+\varepsilon}}{\Gamma(1+\varepsilon)} = \tau \sum_{j\ge 0} \mathcal{E}_j(L)\frac{\varepsilon^j}{j!}, \qquad \mathcal{B}_0(1+\varepsilon; a, \tau) = \mathcal{Q}_{-\varepsilon}(a).$$

The normally convergent lattice series permits coefficient extraction using (4.2). The Leibniz coefficient is $(-1)^m(-1)^{m-j} = (-1)^j$, giving the stated sign. All order extractions on the right are permitted by (2.8). □

4.1 The first three identities

With $\mathcal{Q}_0^{[r]} = \partial_\nu^r \mathcal{Q}_\nu|_{\nu=0}$,

$$\sum_n \left(\operatorname{Log} w_n - \psi(w_n) - \frac{1}{2w_n} \right) = \tau \mathcal{Q}_0, \tag{4.6}$$

$$\sum_n \left(\gamma_1(w_n) + \frac{\operatorname{Log}^2 w_n}{2} - \frac{\operatorname{Log} w_n}{2w_n} \right) = \tau(\mathcal{Q}_0^{[1]} - L\mathcal{Q}_0), \tag{4.7}$$

$$\sum_n \left(\gamma_2(w_n) + \frac{\operatorname{Log}^3 w_n}{3} - \frac{\operatorname{Log}^2 w_n}{2w_n} \right) = \tau(\mathcal{Q}_0^{[2]} - 2L\mathcal{Q}_0^{[1]} + (L^2 - \zeta(2))\mathcal{Q}_0). \tag{4.8}$$

For real a, opposite lattice terms are conjugate; the left sides are real. The right sides are real as well, since $0 < z, q < 1$. For example, at $a = 0.8$ and $h = 2.1$, (4.6) is approximately 0.0690653783784057. This displayed decimal is a diagnostic value, not an interval enclosure.

4.2 Finite horizontal differences remove the Lambert series

For an integer $J \geq 1$, geometric cancellation gives

$$\begin{aligned} \mathcal{Q}_\nu(a) - \mathcal{Q}_\nu(a+J) = \sum_{r=0}^{J-1} \operatorname{Li}_\nu(zq^r) - \frac{1}{2}(\operatorname{Li}_\nu(z) - \operatorname{Li}_\nu(zq^J)) \\ - \frac{1}{\tau}(\operatorname{Li}_{\nu+1}(z) - \operatorname{Li}_{\nu+1}(zq^J)). \end{aligned} \tag{4.9}$$

Thus the difference of the two Stieltjes lattice sums in (4.5) has a *finite* polylogarithmic expression, with all order derivatives and Bell coefficients explicit. In particular, for $J = 1$ the bracket is

$$\frac{1}{2}(\operatorname{Li}_\nu(z) + \operatorname{Li}_\nu(zq)) - \frac{1}{\tau}(\operatorname{Li}_{\nu+1}(z) - \operatorname{Li}_{\nu+1}(zq)). \tag{4.10}$$

Equations (4.5) and (4.10) constitute an infinite family of bilateral generalized-Stieltjes identities with only finitely many polylogarithms on the right. Their derivation uses only absolute sums. They do not make an arithmetic independence claim about the order derivatives that occur.

5 Differentiation and normalized antiderivatives

The Mellin normalization makes horizontal calculus transparent:

$$\partial_w^r \mathcal{R}_M(s, w) = (-1)^r (s)_r \mathcal{R}_M(s + r, w). \tag{5.1}$$

This follows by differentiation of (2.6) in a common convergence domain and then analytic continuation. Accordingly, whenever the original lattice sum lies in its normal domain,

$$\partial_a^r \mathcal{S}_M(s; a, h) = (-1)^r (s)_r \mathcal{S}_M(s + r; a, h). \tag{5.2}$$

Subsequent spectral differentiation is legitimate on every compact subset of that domain. The derivative of the rising factorial is part of the formula and must not be omitted.

There is an equally simple integral form. Since

$$\partial_a \mathcal{Q}_\nu(a) = -\tau \mathcal{Q}_{\nu-1}(a), \tag{5.3}$$

raising the order supplies an anchored primitive.

Theorem 5.1 (All-index anchored primitives)**.** *Let a_0, a lie in the right half-plane and integrate along any path there. Then*

$$\int_{a_0}^{a} \sum_{n\in\mathbb{Z}} K_m(u+\mathrm{i}hn)\,\mathrm{d}u = -\sum_{j=0}^{m} \binom{m}{j} (-1)^j \mathcal{E}_j(L) \cdot \partial_\nu^{m-j}\big(\mathcal{Q}_\nu(a) - \mathcal{Q}_\nu(a_0)\big)\Big|_{\nu=1}. \tag{5.4}$$

In particular, the primitive vanishes at $a = a_0$. More generally,

$$\int_{a_0}^{a} \mathcal{S}_M(s;u,h)\,\mathrm{d}u = -\frac{\tau^{s-1}}{\Gamma(s)}\big(\mathcal{B}_M(s-1;a,\tau) - \mathcal{B}_M(s-1;a_0,\tau)\big) \tag{5.5}$$

on the normal domain $\operatorname{Re} s > -2M$*.*

Proof. Differentiate the right sides using $\partial_a \operatorname{Li}_\nu(e^{-\tau a}) = -\tau \operatorname{Li}_{\nu-1}(e^{-\tau a})$ and its normally convergent Lambert counterpart. This gives the integrands by Theorems 3.2 and 4.1. The expressions vanish at a_0, and the right half-plane is simply connected. Normal convergence justifies the integral of the lattice sum along a compact path. This proof avoids an unjustified attempt to integrate each nonconvergent primitive separately. □

When $\operatorname{Re}(s-1) > -2M$ and $s \neq 1$, (5.5) may equivalently be written

$$\frac{\mathcal{S}_M(s-1;a_0,h) - \mathcal{S}_M(s-1;a,h)}{s-1}.$$

Outside that smaller domain, this last notation refers to the continued summed function, not necessarily an ordinary series of primitives. Formula (5.5) itself remains unambiguous on the larger stated domain.

6 The Gamma specialization and its known product identity

Define the normalized Stirling remainder

$$G(w) = \log\Gamma(w) - (w-\tfrac{1}{2})\operatorname{Log} w + w - \tfrac{1}{2}\log(2\pi). \tag{6.1}$$

The Hurwitz special value $\zeta'(0,w) = \log\Gamma(w) - \frac{1}{2}\log(2\pi)$ implies

$$\partial_s \mathcal{R}_1(0,w) = G(w) - \frac{1}{12w}. \tag{6.2}$$

This is inside the normal domain of $\mathcal{S}_1$. Differentiating (3.2) there gives the absolutely convergent identity

$$\sum_{n\in\mathbb{Z}} \left(G(w_n) - \frac{1}{12w_n}\right) = \mathcal{L}_1(q,z) - \frac{1}{2}\operatorname{Li}_1(z) - \frac{1}{\tau}\operatorname{Li}_2(z) - \frac{\tau}{12}\operatorname{Li}_0(z). \tag{6.3}$$

For a function with reciprocal decay we specify

$$\operatorname{Sym}\sum_{n\in\mathbb{Z}} F(w_n) := \lim_{N\to\infty} \sum_{n=-N}^{N} F(w_n).$$

The elementary partial-fraction identity is

$$\operatorname{Sym}\sum_{n\in\mathbb{Z}} \frac{1}{a+\mathrm{i}hn} = \frac{\tau}{2}\coth\frac{\tau a}{2} = \tau\left(\operatorname{Li}_0(z) + \frac{1}{2}\right). \tag{6.4}$$

For completeness, pair n and $-n$ to obtain $1/a + \sum_{n\geq 1} 2a/(a^2+h^2n^2)$; the standard partial fractions of coth give (6.4). They follow as well by logarithmically differentiating the canonical product for sinh.

Corollary 6.1 (Known Gamma-product specialization)**.** *For $h > 0$ and $\operatorname{Re} a > 0$,*

$$\boxed{\operatorname{Sym} \sum_{n\in\mathbb{Z}} G(a + \mathrm{i}hn) = -\log(z;q)_\infty + \frac{1}{2}\log(1-z) - \frac{1}{\tau}\operatorname{Li}_2(z) + \frac{\tau}{24}.} \tag{6.5}$$

Here $\log(z;q)_\infty = \sum_{r\geq 0}\log(1-zq^r)$, with the analytic logarithm supplied by its convergent series.

Proof. Add one twelfth of (6.4) to (6.3), use $\mathcal{L}_1 = -\log(z;q)_\infty$ and $\operatorname{Li}_1(z) = -\log(1-z)$. The Li_0 terms cancel, leaving $\tau/24$. □

Equation (6.5), after exponentiation and the substitution $y = \tau a$, $\beta = \tau$, is precisely the Gamma-product specialization credited to Arabi Ardehali and Rosengren [7, Eq. (13)]. Its absolute version corresponds to their subtracted remainder. We rederive it only to fix the normalization of the subsequent general contact law; neither the Gamma product nor its constant is claimed as new.

There is a useful interchange diagnostic. Differentiating the entire continuation of $\mathcal{S}_0(s)$ at $s = 0$ gives $\mathcal{B}_0(0)$, whereas symmetrically summing $\partial_s\mathcal{R}_0(0, w_n) = G(w_n)$ gives $\mathcal{B}_0(0) + \tau/24$. The discrepancy is exactly $\tau/24$, not a numerical error. At $a = 0.8$, $h = 2.1$, the correct symmetric value is approximately 0.14613043538142660274.

7 Negative integer jets: a parity-sensitive contact law

For $N \geq 0$, put

$$M_N = \left\lceil \frac{N}{2} \right\rceil, \qquad T_N(w) = \partial_s\mathcal{R}_{M_N}(s, w)\big|_{s=-N}. \tag{7.1}$$

The chosen compensation is minimal for the parity statement below. Every $\mathcal{R}_{M_N}(-N, w)$ is zero. This follows from (2.6), whose integral is convergent there, or directly from the Bernoulli-polynomial formula for $\zeta(-N, w)$.

Proposition 7.1 (Explicit harmonic-number form)**.** *For $N \geq 0$ and $\operatorname{Re} w > 0$,*

$$\begin{aligned} T_N(w) = \zeta'(-N, w) &- \frac{w^{N+1}\operatorname{Log} w}{N+1} + \frac{w^{N+1}}{(N+1)^2} + \frac{1}{2}w^N \operatorname{Log} w \\ &- \sum_{j=1}^{M_N} \frac{B_{2j}}{(2j)!}\frac{N!}{(N-2j+1)!}\, w^{N-2j+1}(\operatorname{Log} w + H_N - H_{N-2j+1}). \end{aligned} \tag{7.2}$$

The sum is empty when $N = 0$.

Proof. For $k = 2j - 1 \leq N$,

$$(-N)_k = -\frac{N!}{(N-k)!}, \qquad \frac{\partial_s(s)_k}{(s)_k}\bigg|_{s=-N} = -H_N + H_{N-k}.$$

Differentiate each term of (2.4). The two minus signs in the derivative of $(s)_k w^{1-s-2j}$ give the displayed harmonic factor. Differentiating the polar subtraction contributes $-w^{N+1}\operatorname{Log} w/(N+1) + w^{N+1}/(N+1)^2$. □

Lemma 7.2 (The first nonzero reciprocal coefficient)**.** *If N is even, then*

$$T_N(w) = \frac{\beta_N}{w} + O(w^{-3}), \qquad \beta_N = \frac{B_{N+2}}{(N+1)(N+2)}. \tag{7.3}$$

If N is odd, then $T_N(w) = O(w^{-2})$. These estimates are uniform on closed sectors avoiding the negative axis.

Proof. For $k \geq N+1$, the product $(s)_k$ has exactly one zero factor at $s=-N$, and

$$\partial_s (s)_k|_{s=-N} = (-1)^N N!(k-N-1)!. \tag{7.4}$$

Differentiate the asymptotic expansion (2.7). No logarithm remains in its terms because $(s)_k$ itself vanishes at $-N$. The first remaining exponent is -1 for even N and -2 for odd N. In the even case its coefficient is (7.3); successive exponents differ by two. □

Theorem 7.3 (All nonpositive integer first jets)**.** *For $N \geq 0$, $h > 0$, and $\operatorname{Re} a > 0$,*

$$\boxed{\operatorname{Sym} \sum_{n\in\mathbb{Z}} T_N(a+\mathrm{i}hn) = (-1)^N N!\tau^{-N}\mathcal{B}_{M_N}(-N;a,\tau) + \mathbf{1}_{2|N}\frac{\tau B_{N+2}}{2(N+1)(N+2)}.} \tag{7.5}$$

For odd N, the sum is absolutely convergent and Sym *may be omitted. For even N, the displayed symmetric prescription exists; the contact is independent of a.*

Proof. When N is odd, $-N > -2M_N$. Normal convergence in Theorem 3.2 permits differentiation at $s=-N$. Since $(1/\Gamma)'(-N) = (-1)^N N!$, this gives the first term on the right and no contact.

Let $N = 2M$ be even. Oversubtract once, using $\mathcal{R}_{M+1}$. The order $-N$ is now strictly inside the normal convergence domain. The difference of the two first jets is exactly β_N/w, by (7.4). Thus

$$\operatorname{Sym} \sum_n T_N(w_n) = N!\tau^{-N}\mathcal{B}_{M+1}(-N;a,\tau) + \beta_N \frac{\tau}{2}\coth\frac{\tau a}{2}.$$

By the definition of $\mathcal{B}_M$,

$$N!\tau^{-N}\mathcal{B}_{M+1}(-N) = N!\tau^{-N}\mathcal{B}_M(-N) - \beta_N \tau \operatorname{Li}_0(z).$$

Use $\coth(\tau a/2) = 1 + 2\operatorname{Li}_0(z)$. The image terms cancel and leave $\beta_N\tau/2$. Existence of the symmetric sum follows from Lemma 7.2 and (6.4). □

7.1 Two explicit higher identities

The first negative odd order has

$$T_1(w) = \zeta'(-1,w) - \tfrac{1}{2}w^2 \operatorname{Log} w + \tfrac{1}{4}w^2 + \tfrac{1}{2}w\operatorname{Log} w - \tfrac{1}{12}(\operatorname{Log} w + 1), \tag{7.6}$$

$$\sum_{n\in\mathbb{Z}} T_1(w_n) = -\frac{\mathcal{L}_2(q,z)}{\tau} + \frac{\operatorname{Li}_2(z)}{2\tau} + \frac{\operatorname{Li}_3(z)}{\tau^2} + \frac{\operatorname{Li}_1(z)}{12}. \tag{7.7}$$

This is an ordinary absolutely convergent identity. The next order is

$$T_2(w) = \zeta'(-2,w) - \tfrac{1}{3}w^3 \operatorname{Log} w + \tfrac{1}{9}w^3 + \tfrac{1}{2}w^2\operatorname{Log} w - \tfrac{1}{6}w\operatorname{Log} w - \tfrac{1}{12}w, \tag{7.8}$$

$$\operatorname{Sym} \sum_{n\in\mathbb{Z}} T_2(w_n) = \frac{2}{\tau^2}\mathcal{L}_3(q,z) - \frac{\operatorname{Li}_3(z)}{\tau^2} - \frac{2\operatorname{Li}_4(z)}{\tau^3} - \frac{\operatorname{Li}_2(z)}{6\tau} - \frac{\tau}{720}. \tag{7.9}$$

The final $-\tau/720$ is required. These examples display explicitly how negative-order Hurwitz derivatives connect to polylogarithms of successive positive integer orders. Formula (7.2) provides the corresponding harmonic-number expression at every order.

7.2 Periodic filters annihilate the mean contact

Corollary 7.4 (Mean-zero periodic weights)**.** *Let c_n be periodic of period $d \geq 1$. Put $\tau_d = \tau/d$ and $a_r = a + \mathrm{i}hr$ for $0 \leq r < d$. Then*

$$\begin{aligned}\operatorname{Sym} \sum_{n\in\mathbb{Z}} c_n T_N(a+\mathrm{i}hn) = (-1)^N N!\tau_d^{-N} \sum_{r=0}^{d-1} c_r \mathcal{B}_{M_N}(-N; a_r, \tau_d) \\ + \mathbf{1}_{2|N} \frac{\tau_d B_{N+2}}{2(N+1)(N+2)} \sum_{r=0}^{d-1} c_r.\end{aligned} \tag{7.10}$$

Consequently every first-jet contact vanishes for a mean-zero filter.

Proof. Split the lattice into residue classes modulo d and apply Theorem 7.3 to each class with spacing dh. For even N, shifting the endpoints of a symmetric cutoff by a bounded number of terms changes its limit by zero, because $T_N(w_n) = O(1/|n|)$. For odd N regrouping is absolute. The contact in each class is the same constant. Summing yields (7.10). □

For example, the alternating Gamma specialization is

$$\operatorname{Sym} \sum_{n\in\mathbb{Z}} (-1)^n G(a+\mathrm{i}hn) = 2\mathcal{B}_0(0; a, \tau/2) - \mathcal{B}_0(0; a, \tau). \tag{7.11}$$

Indeed the even-index lattice has spacing $2h$, and the two contacts cancel. Formula (7.10) identifies the obstruction as a single mean mode, not an arbitrary choice of regularization.

8 A harmonic-number compiler for every higher spectral jet

At a negative integer, reciprocal Gamma has the exact local factorization

$$\frac{\tau^{-N+\varepsilon}}{\Gamma(-N+\varepsilon)} = (-1)^N N!\tau^{-N}\varepsilon \, \frac{\tau^{\varepsilon}}{\Gamma(1+\varepsilon)} \prod_{k=1}^{N} \left(1 - \frac{\varepsilon}{k}\right). \tag{8.1}$$

Define $\mathcal{E}_j^{(N)}$ by

$$\begin{aligned}\exp\left((\log\tau + \gamma - H_N)u + \sum_{r=2}^{\infty} \frac{(-1)^{r+1}\zeta(r) - H_N^{(r)}}{r} u^r\right) \\ = \sum_{j=0}^{\infty} \mathcal{E}_j^{(N)} \frac{u^j}{j!}.\end{aligned} \tag{8.2}$$

For $L_N = \log\tau + \gamma - H_N$,

$$\begin{aligned}\mathcal{E}_0^{(N)} &= 1, \qquad \mathcal{E}_1^{(N)} = L_N, \\ \mathcal{E}_2^{(N)} &= L_N^2 - \zeta(2) - H_N^{(2)}, \\ \mathcal{E}_3^{(N)} &= L_N^3 - 3L_N(\zeta(2) + H_N^{(2)}) + 2\zeta(3) - 2H_N^{(3)}.\end{aligned}$$

These formulas keep the generalized harmonic correction separate from the alternating sign of the zeta contribution.

Theorem 8.1 (Every compensated negative-integer jet)**.** *Let $N \geq 0$, L be an integer with $L > N/2$, and $d \geq 1$. Then the following lattice sum is absolutely convergent:*

$$\begin{aligned}&\sum_{n\in\mathbb{Z}} \partial_s^d \mathcal{R}_L(s, a + \mathrm{i}hn)\Big|_{s=-N} \\ &\quad = (-1)^N N!\tau^{-N} d \sum_{j=0}^{d-1} \binom{d-1}{j} \mathcal{E}_j^{(N)}\, \partial_s^{d-1-j} \mathcal{B}_L(s; a, \tau)\Big|_{s=-N}.\end{aligned} \tag{8.3}$$

Every derivative on the right is an explicit polylogarithm order derivative or a Lambert sum of such derivatives.

Proof. The strict inequality $L > N/2$ places $-N$ inside the normal convergence domain of Theorem 3.2. Expand its right side using (8.1). Taking the logarithm of the finite product gives $-\sum_{r\geq 1} H_N^{(r)} \varepsilon^r / r$, which, together with the reciprocal-Gamma series, yields (8.2). The coefficient of ε^d in $\varepsilon \mathcal{E}^{(N)}(\varepsilon) \mathcal{B}_L(-N + \varepsilon)$ is a Leibniz coefficient of degree $d - 1$. Multiplication by $d!$ gives exactly the factor $d\binom{d-1}{j}$ in (8.3). □

This theorem is a finite symbolic procedure at every prescribed (N, d, L). It involves no numerical guessing of coefficients. For the minimally compensated even-order first jet one may instead use Theorem 7.3; for higher minimal jets, logarithmic reciprocal tails generally appear. No ordinary-sum interpretation of those insufficiently compensated higher jets is asserted here.

9 Products of polygammas: finite Hurwitz–Lerch images

In this section $a, b > 0$ are real. Let $r, t \geq 2$ be integers, set $A = a + b$, and define

$$D_\ell(Q, A) = \frac{\zeta(\ell, A) - Q\Phi(Q, \ell, A)}{1 - Q}, \qquad 0 < Q < 1, \quad \ell \geq 2, \tag{9.1}$$

where

$$\Phi(Q, \ell, A) = \sum_{m=0}^{\infty} \frac{Q^m}{(m + A)^\ell}.$$

The elementary partial fractions

$$\frac{1}{(1 - x)(1 - Qx)} = \frac{1}{1 - Q}\left(\frac{1}{1 - x} - \frac{Q}{1 - Qx}\right) \tag{9.2}$$

give the convergent integral

$$\int_0^\infty \frac{u^{\ell-1} e^{-Au}}{(1 - e^{-u})(1 - Qe^{-u})}\, \mathrm{d}u = \Gamma(\ell) D_\ell(Q, A). \tag{9.3}$$

Theorem 9.1 (Finite-image product identity)**.** *For $a, b, h > 0$ and integers $r, t \geq 2$,*

$$\sum_{n\in\mathbb{Z}} \zeta(r, a + \mathrm{i}hn)\zeta(t, b - \mathrm{i}hn) = V_{r,t}(A) + \sum_{k=1}^{\infty} \big(I_{r,t;k}(a, b) + I_{t,r;k}(b, a)\big), \tag{9.4}$$

where

$$V_{r,t}(A) = \frac{\tau\Gamma(r + t - 1)}{\Gamma(r)\Gamma(t)} \big(\zeta(r + t - 2, A) - (A - 1)\zeta(r + t - 1, A)\big), \tag{9.5}$$

$$I_{r,t;k}(a, b) = \frac{\tau q^{ak}}{\Gamma(r)\Gamma(t)} \sum_{j=0}^{r-1} \binom{r-1}{j} (\tau k)^{r-1-j} \Gamma(j + t) D_{j+t}(q^k, A). \tag{9.6}$$

The lattice sum is absolutely convergent. The image series converges exponentially, and each individual image is a finite combination of Hurwitz zeta and Lerch values.

Proof. Extend the two functions

$$f(u) = \frac{u^{r-1}e^{-au}}{\Gamma(r)(1-e^{-u})}, \qquad g(u) = \frac{u^{t-1}e^{-bu}}{\Gamma(t)(1-e^{-u})}$$

by zero on the negative axis. They are in $L^1 \cap L^2$. Their Fourier transforms are $\zeta(r, a+\mathrm{i}y)$ and $\zeta(t, b+\mathrm{i}y)$. The correlation

$$P(x) = \int_{\mathbb{R}} f(v+x)g(v)\,\mathrm{d}v$$

is continuous by continuity of translations in L^2, decays exponentially up to polynomial factors in both directions, and has Fourier transform $\widehat{f}(y)\widehat{g}(-y)$. The sampled transform is absolutely summable since the product is $O(|n|^{2-r-t})$. Apply Lemma 3.1 to P.

At $x = 0$ expand $(1-e^{-u})^{-2} = \sum_{m\geq 0}(m+1)e^{-mu}$. Integration gives (9.5), since $m+1 = (m+A)-(A-1)$. For $x = \tau k > 0$ expand $(v+\tau k)^{r-1}$ by the binomial theorem and apply (9.3). This gives (9.6). Negative x swaps (r, a) and (t, b). Exponential decay of the images follows also directly from the integral, with factors q^{ak} or q^{bk} and a fixed polynomial in k. □

The usual relation

$$\zeta(r, w) = \frac{(-1)^r}{(r-1)!}\psi^{(r-1)}(w) \tag{9.7}$$

converts the theorem into a formula for every product of two positive polygamma derivatives. It also permits a finite harmonic-number interpretation after horizontal integer differences are taken.

9.1 An explicit trigamma square

For $a > 0$, $r = t = 2$ and $b = a$ give

$$\begin{aligned}\sum_{n\in\mathbb{Z}} |\psi'(a+\mathrm{i}hn)|^2 &= 2\tau\big(\zeta(2,2a) - (2a-1)\zeta(3,2a)\big) \\ &\quad + 2\tau\sum_{k=1}^{\infty} q^{ak}\big(\tau k D_2(q^k, 2a) + 2D_3(q^k, 2a)\big).\end{aligned} \tag{9.8}$$

The $k = 0$ term is the continuous vertical pairing at the sampling normalization. Every $k \geq 1$ term is a discrete image correction. Keeping only the continuous term is therefore not an exact lattice identity.

9.2 Finite generalized-harmonic differences

The shift formula

$$\zeta(r, w) - \zeta(r, w+J) = \sum_{j=0}^{J-1}(w+j)^{-r}$$

turns four horizontal evaluations of (9.4) into an exact pairing of finite generalized-harmonic sums. All finite sums may be interchanged with the lattice series because its terms decay at least quadratically. Thus the same identity system covers both infinite polygamma tails and finite harmonic windows; no separate regularization is needed.

10 All-index Stieltjes differences and four-shift closure

Let $a, c, b, d > 0$ and write

$$\Delta_{a,c}(s; n) = \zeta(s, a + \mathrm{i}hn) - \zeta(s, c + \mathrm{i}hn), \tag{10.1}$$

$$\delta_m^{a,c}(n) = \gamma_m(a + \mathrm{i}hn) - \gamma_m(c + \mathrm{i}hn). \tag{10.2}$$

The pole at $s = 1$ cancels in Δ, and

$$\Delta_{a,c}(1 + \varepsilon; n) = \sum_{m \geq 0} \frac{(-1)^m}{m!} \delta_m^{a,c}(n) \varepsilon^m.$$

For each fixed m, $\delta_m^{a,c}(n) = O(\log^m(2 + |n|)/(1 + |n|))$. Its product with a second such difference is absolutely summable.

For $\operatorname{Re} s > 0$ set

$$f_s^{a,c}(u) = \frac{u^{s-1}}{\Gamma(s)} \frac{e^{-au} - e^{-cu}}{1 - e^{-u}}, \qquad u > 0, \tag{10.3}$$

extended by zero to $u < 0$. It belongs to L^1; when $\operatorname{Re} s > 1/2$ it also belongs to L^2. Its Fourier transform is $\Delta_{a,c}(s; y/h)$ in the evident continuous-variable notation.

Theorem 10.1 (Difference pairing and all its Stieltjes coefficients)**.** *For* $\operatorname{Re} s, \operatorname{Re} t > 1/2$*,*

$$\begin{aligned} &\sum_{n \in \mathbb{Z}} \Delta_{a,c}(s; n) \Delta_{b,d}(t; -n) \\ &= \tau \int_0^\infty f_s^{a,c}(u) f_t^{b,d}(u) \, \mathrm{d}u \\ &\quad + \tau \sum_{k=1}^\infty \int_0^\infty \left(f_s^{a,c}(u + \tau k) f_t^{b,d}(u) + f_s^{a,c}(u) f_t^{b,d}(u + \tau k) \right) \mathrm{d}u. \end{aligned} \tag{10.4}$$

All series, integrals, and fixed spectral derivatives converge normally near $(s, t) = (1, 1)$*. Define*

$$F_m^{a,c}(u) = (-1)^m \mathcal{E}_m(\log u + \gamma) \frac{e^{-au} - e^{-cu}}{1 - e^{-u}}. \tag{10.5}$$

Then, for every $m, \ell \geq 0$*,*

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \delta_m^{a,c}(n) \delta_\ell^{b,d}(-n) &= \tau \int_0^\infty F_m^{a,c}(u) F_\ell^{b,d}(u) \, \mathrm{d}u \\ &\quad + \tau \sum_{k \geq 1} \int_0^\infty \left(F_m^{a,c}(u + \tau k) F_\ell^{b,d}(u) \right. \\ &\qquad \left. + F_m^{a,c}(u) F_\ell^{b,d}(u + \tau k) \right) \mathrm{d}u. \end{aligned} \tag{10.6}$$

These are ordinary absolutely convergent identities.

Proof. Use the correlation and periodization argument of Theorem 9.1, now for the $L^1 \cap L^2$ densities (10.3). The shift difference improves the vertical Hurwitz decay to $O(|n|^{-\operatorname{Re} s})$, so the sampled product is absolutely summable in the stated domain. At zero, the densities behave as a constant times u^{s-1}; at infinity they decay exponentially. These facts give the required normal convergence, including logarithmic factors from spectral derivatives.

The coefficient identity $(-1)^m \partial_s^m f_s^{a,c}|_{s=1} = F_m^{a,c}$ follows from (4.3). Extract the two coefficients in (10.4). The signs and factorials then match (10.6). □

10.1 The central image has finite Hurwitz–Stieltjes coordinates

Let $a_1 = a$, $a_2 = c$, $b_1 = b$, $b_2 = d$, and $\epsilon_1 = 1$, $\epsilon_2 = -1$. Put

$$\mathcal{H}(v) = \sum_{i,j=1}^{2} \epsilon_i\epsilon_j\big(\zeta(v-1, a_i+b_j) - (a_i+b_j-1)\zeta(v, a_i+b_j)\big). \tag{10.7}$$

The potential poles at $v = 2$ and $v = 1$ cancel because

$$\sum_{i,j} \epsilon_i\epsilon_j = 0, \qquad \sum_{i,j} \epsilon_i\epsilon_j(a_i+b_j-1) = 0.$$

Hence $\mathcal{H}$ is entire. The central term of (10.4) is

$$\tau\frac{\Gamma(s+t-1)}{\Gamma(s)\Gamma(t)}\mathcal{H}(s+t-1). \tag{10.8}$$

To prove it, expand the squared geometric denominator in a common absolute-convergence domain and integrate termwise, then continue the grouped expression to $\operatorname{Re} s, \operatorname{Re} t > 1/2$.

In particular,

$$\mathcal{H}^{(k)}(1) = \sum_{i,j} \epsilon_i\epsilon_j \left(\zeta^{(k)}(0, a_i+b_j) - (-1)^k(a_i+b_j-1)\gamma_k(a_i+b_j)\right). \tag{10.9}$$

Every central Stieltjes coefficient is therefore a finite combination of these values and reciprocal-Gamma Bell coefficients. Explicitly, the central contribution to (10.6) is

$$\tau(-1)^{m+\ell}m!\ell![u^m v^\ell]\frac{\Gamma(1+u+v)}{\Gamma(1+u)\Gamma(1+v)}\mathcal{H}(1+u+v). \tag{10.10}$$

The nonzero images in (10.6) remain explicit one-dimensional logarithmic integrals. We do not claim a finite single-Hurwitz closure for arbitrary spectral derivatives of those images.

10.2 A fully explicit four-shift digamma identity

Theorem 10.2 (Four-shift closure)**.** *With the preceding positive real parameters and $A_{ij} = a_i + b_j$, define*

$$\mathcal{P} = \sum_{n\in\mathbb{Z}} \big(\psi(c+\mathrm{i}hn) - \psi(a+\mathrm{i}hn)\big)\big(\psi(d-\mathrm{i}hn) - \psi(b-\mathrm{i}hn)\big).$$

Then

$$\begin{aligned}\mathcal{P} = {}& \tau\sum_{i,j}\epsilon_i\epsilon_j(A_{ij}-1)\psi(A_{ij}) \\ & - \tau\sum_{k=1}^{\infty}\frac{1}{1-q^k}\sum_{i,j}\epsilon_i\epsilon_j\big(q^{a_i k}+q^{b_j k}\big) \\ & \qquad \cdot\big(\psi(A_{ij}) + q^k\Phi(q^k, 1, A_{ij})\big).\end{aligned} \tag{10.11}$$

The original lattice series is absolutely convergent and the image series converges exponentially.

Proof. Put $s = t = 1$ in (10.4). The central image is $\tau\mathcal{H}(1)$; the $\zeta(0, A)$ terms in (10.9) cancel, giving the first line of (10.11). For a positive image, use (9.2). Its would-be Hurwitz order-one pole is independent of A_{ij}, and its total coefficient is

$$\sum_{i,j}\epsilon_i\epsilon_j q^{a_i k} = 0.$$

Thus the finite part of the grouped expression uses $\zeta(1+\eta, A) = \eta^{-1} - \psi(A) + O(\eta)$ and gives

$$-\frac{\tau}{1-q^k} \sum_{i,j} \epsilon_i \epsilon_j q^{a_i k} \big(\psi(A_{ij}) + q^k \Phi(q^k, 1, A_{ij})\big).$$

This is an ordinary convergent grouped integral; the Laurent calculation only evaluates it. The negative image has $q^{b_j k}$ instead. Adding them proves (10.11). □

Example 10.3 (A normalization check that reduces to elementary functions). If $c = a + 1$ and $d = b + 1$, the two digamma differences are reciprocals. Consequently (10.11) must reduce to

$$\sum_{n\in\mathbb{Z}} \frac{1}{(a + \mathrm{i}hn)(b - \mathrm{i}hn)} = \frac{\tau}{2(a+b)} \left(\coth \frac{\tau a}{2} + \coth \frac{\tau b}{2} \right). \tag{10.12}$$

This follows independently by partial fractions and (6.4). The paired product is itself absolutely convergent. Longer integer shift windows reduce to finite sums of (10.12), giving further independent checks of all signs and scale factors in the four-shift image formula.

11 Verification, corrections, and integration status

11.1 What was checked

The distributed `verification/run_checks.py` executes 268 exact algebraic assertions and 38 floating-point comparisons at 65 working decimal digits. All passed in the recorded run. The largest recorded scaled residual was approximately 3.55×10^{-43}; tolerances were 10^{-32} or 10^{-30}, depending on the test. These residuals are diagnostics, not rigorous error bounds.

The exact assertions verify Bernoulli-polynomial cancellation at negative integer orders, the zero-factor derivative (7.4), the first contact coefficient, harmonic-number factors in (7.2), and the rational partial fractions used by the image formulas. They test finitely many instances of the uniform proofs; they are not presented as a substitute for those proofs.

The numerical routes are deliberately separated. The lattice side uses direct Hurwitz evaluations for a finite central range, then a sectorial Euler–Maclaurin tail summed through Hurwitz zeta values. The Lambert side uses rapidly convergent real-exponential series. Stieltjes coefficients are obtained from a pole-cancelled, complex-safe Euler–Maclaurin expression rather than a call to `mpmath.stieltjes` at nonreal arguments. Integer product pairings use independently convolved large-lattice asymptotic coefficients. Four-shift tests include finite reciprocal reductions and direct integral checks of noninteger-shift images.

Some checks compare two symbolic consequences of the same master identity, notably the higher-jet Bell compiler and primitive transport; these are labeled as cross-route checks, not as fully independent validations of the master theorem. Two corruption controls reject a changed zero-factor derivative and omission of the Gamma contact. Complete numerical values, tolerances, versions, and classifications are recorded in `results/verification.json`.

11.2 A concrete editorial correction

The inspected manuscript README contains the same paragraph beginning “The complete 1,032-input first-letter X family” twice in immediate succession [8]. The proposed correction is to retain one copy and delete the duplicate, without changing any mathematical status statement. This is an editorial correction, not evidence of a false theorem. The package contains a guarded script that removes the second occurrence only when the exact duplicated paragraph is present in a supplied local README copy. It does not contact or modify GitHub.

11.3 A mathematical warning, not an attributed repository error

The tempting identity

$$\operatorname{Sym}\sum_n G(w_n) \overset{\text{wrong}}{=} \frac{\partial}{\partial s}\frac{\tau^s \mathcal{B}_0(s)}{\Gamma(s)}\Big|_{s=0}$$

omits $\tau/24$. More generally, the omitted term at even negative order $-N$ is the contact in (7.5). We provide this as an explicit correction to that invalid interchange. The audit did *not* establish that an inspected canonical theorem made this mistake, and it should not be recorded as an error in a particular existing manuscript without an actual source occurrence.

11.4 Placement and proof status

A natural placement is a new continuation under the existing vertical-pairing report. The core depends only on the standard Hurwitz/Mellin identities and the elementary periodization lemma. It does not require accepting any pending continuum or arithmetic claim in an incoming archive. A proposed integration order is: compensation and sampling; Stieltjes jets and primitives; negative-integer contacts; higher jets; integer pairings; Stieltjes differences and four-shift closure.

All theorem statements in this article are supplied with analytic proofs. No proof-assistant formalization is claimed. The basic Gamma specialization is known and credited. The general harmonic-prefix conjecture, S_6, and revised S_8 remain outside the proved scope of this package. Formal coefficient calculations made during the source search do not settle those period identities. The original incoming archives and manuscript files are left unchanged.

12 Further research questions

1. Minimal compensation for higher resonant jets. The first-jet contact is a single Bernoulli constant. For $\partial_s^d \mathcal{R}_{N/2}(-N, w)$ with even N and $d \geq 2$, logarithmic reciprocal tails appear. Determine a fixed symmetric cutoff subtraction that evaluates these minimal jets, and derive the complete contact polynomial in $\log\tau$ and the cutoff normalization. Theorem 8.1 supplies an ordinary oversubtracted reference against which every proposed subtraction can be tested.

2. Rationally related vertical scales. Replace $\zeta(r, a+\mathrm{i}hn)\zeta(t, b-\mathrm{i}hn)$ by $\zeta(r, a+\mathrm{i}h_1 n)\zeta(t, b-\mathrm{i}h_2 n)$ with $h_1/h_2 \in \mathbb{Q}$. The correlation constraint is then oblique. Can each sampling image still be reduced to a finite Hurwitz–Lerch grid, uniformly in both integer orders? The continuous rational-scale results in [9] suggest a precise comparison, but do not supply the discrete image terms automatically.

3. Bilinear critical-order contacts. The difference-pairing proof uses $\operatorname{Re} s, \operatorname{Re} t > 1/2$. Study continuation to $s + t = 1$ with the pole cancellation and the lattice cutoff fixed. Determine whether a finite local functional at the Mellin origin controls the discrepancy, as in Theorem 7.3, and identify its complete mixed jet.

4. Higher products and image geometry. For three or more vertical factors, Poisson sampling constrains a sum of Mellin variables to an integer multiple of τ. Determine finite formulas for integer orders on each resulting simplex, and identify when the result closes in multiple Lerch functions or ordinary multiple polylogarithms. A useful first case is a product of two trigammas against one reflected trigamma.

5. Order derivatives of nonzero images. Equation (10.6) is explicit but retains logarithmic integrals. Determine which combinations of mixed spectral jets reduce to finitely many Lerch order derivatives, and which naturally require a genuinely larger function class. A failure of a numerical basis search would not prove nonreducibility.

6. Periodic filters beyond the mean mode. The first resonant contact is killed exactly by a zero-mean periodic filter. For higher minimal jets, determine which additional discrete moments of the filter kill the logarithmic contacts. Relate those conditions to root-of-unity polylogarithm coordinates without asserting arithmetic independence of the resulting values.

7. Modular transport of the Stieltjes jet family. The $a = 1$ Gamma specialization is related to the eta transformation in [7]. Investigate whether selected spectral order derivatives of (3.2) admit a useful exact $\tau \leftrightarrow 4\pi^2/\tau$ transport, including every nonmodular correction. This should begin with a functional identity, not a guessed numerical relation among special values.

8. Formal verification of the analytic interchange. A proof-assistant development could isolate the periodization lemma, normal convergence of compensated Hurwitz jets, and the one-extra-subtraction contact argument. These analytic dependencies are more consequential than a large list of special-value checks. The elementary Bell compiler is a separate, comparatively small formal algebra problem.

13 Conclusion

The discrete vertical problem admits a single organizing mechanism: remove the finitely many Mellin-origin terms that obstruct ordinary summation, periodize the remaining density, and extract spectral coefficients only on a normal convergence domain. This yields exact Stieltjes sums, finite polylogarithmic differences, and normalized primitives. At the first negative-even boundary jet, restoring one removed reciprocal term produces an explicit Bernoulli contact. The same periodization argument applied to correlations gives the finite-image polygamma and four-shift formulas.

The separation between absolute sums, symmetric limits, and continued spectral expressions is part of each identity, not an optional interpretive convention. It is also what makes the formulas usable as independently testable additions to the ProveIt manuscript rather than numerical conjectures.

A Reproduction and source map

The package is self-contained. Compile the article with

```
python build.py
```

and replay the mathematics with

```
python -m pip install -r requirements.txt
python verification/run_checks.py
```

The build script runs three LaTeX passes and checks for undefined references and overfull boxes. The delivered PDF additionally received a rendered layout review. A rebuild may differ in binary metadata; its bytes are not covered by the original checksum.

The machine-readable source audit records the individual paths and blob identifiers returned during inspection. In particular, the continuous vertical report was read at commit `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`; other reads used `main` and returned their own blob identifiers. The paper's source map must not be interpreted as a complete checkout or an audit of unseen ZIP members.

References

[1] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. Definitions, Euler–Maclaurin formulas, special values, and sectorial asymptotics. https://dlmf.nist.gov/25.11. Accessed 11 October 2026.

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[7] A. Arabi Ardehali and H. Rosengren, *A New Product Formula for $(z;q)_\infty$, with Applications to Asymptotics*, Constructive Approximation, published 19 September 2026. DOI: 10.1007/s00365-026-09783-2. https://doi.org/10.1007/s00365-026-09783-2. In particular, Eqs. (9), (13), and the discussion of symmetric products.

[8] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, collective manuscript README, repository `VladimirReshetnikov/ProveIt`, path `Analysis/Polylogarithms/docs/manuscript/README.md`. Inspected blob `078716656e8d73e98415b927cca1a1bfea059134`. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/manuscript.

[9] ProveIt research report, *Vertical-Line Hurwitz Pairings and Exact Stieltjes–Gamma Calculus*, 10 October 2026, README and stated scope. Path under `Analysis/Polylogarithms/docs/reports/`: `stieltjes-correlation-calculus/continuations/vertical-stieltjes-pairings/README.md`. Read at commit `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`; blob `022b7ad4b25e28fd36322dd8ad36edfcf0d36285`.

[10] ProveIt research report, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, 11 October 2026, README and `sections/03b_general_hankel.tex`. Path under `Analysis/Polylogarithms/docs/reports/`: `stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`. Inspected README blob `395bdbb9469f09109b3dad1d1a86822fe2339ac7`. The already proved smaller-order-two reconstruction is not claimed here.